BEGINNER RUN GUIDE

Suicide-Risk Detection Using NLP-Based Text and Speech Analysis

> **Current verified status:** Project implementation complete as a research prototype. Final automated verification: **56/56 checks PASS, 0 failed**.

This document is written for a person who does not know Python, Streamlit, machine learning, or this project.

1. Project location

Main project:

```
C:\Users\RAMESH\Desktop\Projects\Thanmayi 8000
```

Backup before accuracy enhancement:

```
C:\Users\RAMESH\Desktop\Projects\Thanmayi 8000_BACKUP_BEFORE_ACCURACY_TUNING_2026-10-31
```

Copy the entire project folder if moving to another computer.

2. What the project does

The application supports:

- **Text only**
- **Speech only**
- **Text + Speech**

Primary text branch

The current production/demo model uses:

1. frozen BERT embeddings for offline label-correction analysis,
2. UMAP dimensionality reduction,
3. Gaussian Mixture Model clustering,
4. confidence threshold τ = 0.90 to generate corrected SDCNL labels,
5. fast word TF-IDF (1–2 grams) + character TF-IDF (3–5 grams),
6. calibrated LinearSVC for live text inference.

Primary label-corrected evaluation:

- Accuracy: **85.75%**
- Precision: **93.98%**
- Recall: **78.00%**
- F1: **85.25%**

- ROC-AUC: **95.04%**

A separate offline BERT label-corrected research experiment reached **93.14% accuracy**.

> **Important:** The standard original-web-label SDCNL baseline is still preserved separately: TF-IDF + calibrated Linear SVM = 72.56% accuracy. The 85.75% result is a SDCNL-style **label-corrected evaluation**, not the same evaluation target.

Speech branch

The speech branch uses EATD acoustic features and Logistic Regression:

- Accuracy: 63.29%
- Balanced Accuracy: 63.44%
- ROC-AUC: 59.22%

EATD is a depression/distress corpus, not a direct suicide-risk-labelled speech dataset.

3. Required software

Python — REQUIRED

Recommended: **Python 3.11.x**

Official download: https://www.python.org/downloads/

Windows downloads: https://www.python.org/downloads/windows/

During installation, enable **Add Python to PATH** if shown.

Check installation:

```
python --version
```

Browser — REQUIRED

Use Chrome or Microsoft Edge.

VS Code — OPTIONAL

https://code.visualstudio.com/

Git — OPTIONAL

https://git-scm.com/install/windows

FFmpeg — OPTIONAL

For normal demo, use WAV audio. FFmpeg is only needed if another audio format cannot be decoded.
https://ffmpeg.org/

4. First-time setup

Open the project folder and double-click.

```
SETUP_PROJECT.bat
```

It installs all packages from:

```
requirements.txt
```

Important packages include:

- pandas
- numpy
- scikit-learn
- joblib
- librosa
- soundfile
- streamlit
- torch
- transformers
- matplotlib
- flask
- **umap-learn 0.5.12**

Manual installation:

```
cd /d "C:\Users\RAMESH\Desktop\Projects\Thanmayi 8000"
python -m pip install -r requirements.txt
```

5. Verify before demo

Double-click:

```
VERIFY_PROJECT.bat
```

Expected current result:

```
Passed: 56
Failed: 0
Overall: PASS
```

The checks verify enhanced text artifacts, baseline metrics, label-correction reports, speech model, diagrams, text inference, real WAV speech inference, and fusion.

6. Start the application

Double-click:

```
START_PROJECT.bat
```

Or manually:

```
cd /d "C:\Users\RAMESH\Desktop\Projects\Thanmayi 8000"
python -m streamlit run app.py
```

Open: http://localhost:8501

7. Text-only demo

1. Select **Text only**.
2. Enter text.
3. Click **Analyze**.
4. View P_text and screening level.

Verified example:

```
I am feeling very stressed and overwhelmed today.
```

Current enhanced model output is approximately:

- P_text ≈ **0.988**
- Screening level: **High**

The probability changes when the input changes.

8. Speech-only demo

1. Select **Speech only**.
2. Upload a WAV file.
3. Click Analyze.

Verified sample:

```
Datasets\validation\10\negative_out.wav
```

Verified speech probability is approximately:

- P_speech ≈ **0.282**

9. Text + Speech demo

Use the same example text + sample WAV.

Approximate verified values:

- P_text ≈ 0.988
- P_speech ≈ 0.282
- Combined score ≈ **0.776**
- Screening level: **High**

Fusion formula:

```
Combined = 0.70 × P_text + 0.30 × P_speech
```

Bands:

- Low: below 0.40
- Moderate: 0.40 to below 0.70
- High: 0.70 or above

10. Important files

- app.py — Streamlit UI
- src/inference_pipeline.py — live inference
- src/sdcnl_label_correction_experiment.py — offline BERT → UMAP → GMM label correction
- src/finalize_enhanced_fast_text_model.py — final fast enhanced text model
- models/enhanced_text_model/ — primary text model artifacts
- models/text_model/ — standard baseline artifacts
- models/speech_model/ — speech artifacts
- config/fusion_config.json — prototype decision settings
- tests/run_project_tests.py — automated verification
- diagrams/Thanmayi_Final_System_Design_Diagrams.drawio — diagrams
- Thanmayi_Final_Project_Report.pdf — final report

11. Do not retrain for normal demo

For a normal college demonstration, do not rerun:

- UMAP/GMM label correction
- BERT embedding generation
- model tuning
- speech training
- dataset preprocessing

The trained artifacts are already saved.

12. Common problems

Python not recognized

Install Python 3.11 and reopen Command Prompt.

No module named streamlit / umap / sklearn

Run:

```
python -m pip install -r requirements.txt
```

Port 8501 already in use

Use:

```
python -m streamlit run app.py --server.port 8502
```

Then open http://localhost:8502

Audio error

Use WAV first.

MemoryError

Close training scripts and unnecessary applications. Normal demo uses the fast enhanced model and does not need BERT inference.

13. Viva explanation

Say clearly:

“Our primary text model achieves 85.75% accuracy in the SDCNL-style label-corrected evaluation. We preserve the standard original-web-label baseline of 72.56% separately. Label correction uses BERT embeddings, UMAP and GMM with confidence threshold 0.90. The live application uses a fast word+character TF-IDF calibrated LinearSVC trained on the corrected labels.”

Also say:

- Speech uses EATD depression/distress labels.
- SDCNL and EATD are not participant-matched.
- Combined score is a research prototype, not a clinical diagnosis.
- No clinical multimodal accuracy is claimed.

14. Final checklist

- Dataset audit: COMPLETE
- Leakage check: COMPLETE
- Standard text baselines: COMPLETE
- SDCNL label correction: COMPLETE
- Enhanced fast text model: COMPLETE
- Primary corrected accuracy: **85.75%**
- Offline corrected research experiment: **93.14%**
- Speech classifier: COMPLETE
- Text / Speech / Both inference: COMPLETE
- Streamlit UI: COMPLETE
- Six diagrams: PASS
- Diagram ZIP: PASS
- Automated verification: **56/56 PASS**
- Final DOCX/PDF report: COMPLETE

> **FINAL STATUS: COMPLETE AS AN IMPLEMENTATION-READY RESEARCH PROTOTYPE.**